Guided lab 3

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Exercise 4

Description of each line of code:
coeffs[], is used to create an array that will include the coefficients of the piecewise spline to be evaluated.
for the for loop: each row k will have the corresponding coefficient for the k-th polynomial piece, while the coefficients are stored in descending order. This order is needed for the mkpp command.
breaks = [x], this array will contain all the x values that the polynomial will check.
pp = mkpp(breaks, coeffs), this function builds a piecewise polynomial from breaks and coeffs, it essentially divides the interval,so for example, breaks(1) and breaks(2) with the first coefficient (from last) will be the first polynomial piece.
ystar = ppval(pp,xstar), this function is used to evaluate the piecewise polynomial at xstar.

Exercise 5

Using the data from the Bradie text "Mean Activity Coefficient of Silver Nitrate" to test our code, we get this output:
"The estimated activity coefficient at molality 0.032 is: 0.828346"
It is roughly equal to the amount we are supposed to find; however, we are using different boundary conditions. The example from the textbook uses the "not-a-knot" condition, while we are using the natural boundary conditions.

Exercise 6

(a)Estimating the mean weight of a larva three days after birth using the interpolator we built (natural boundary condition) and Matlab's built-in spline function ("not-a-knot" boundary condition), this is the output:
natural bounds young = 7.19, matlab young = -2.71
natural bounds mature = 11.56, matlab mature = 10.45

The results look very different as a result of the nature of these boundary conditions. For the "not-a-knot" condition, we know that $s_o'''(x) = s_1'''(x)$, which means that the first two segments must have the same curve. For the natural boundary condition, we know that $s''(x_o) = s''(x_n) = 0$, which forces the ends to be flat lines. This condition helps to avoid any oscillations since there is no curvature at the end and to obtain results that are within the logical range (young mean = 7.19 and mature mean = 11.56). On the other hand, Matlab's function gives us a negative value for the young mean and a normal value for the mature mean. From day 6 to day 10, there is a major shift in the average weight of the young leaves, so to fit the data, the polynomial forces a large jump, which causes the negative value since it fails to reach the curve that goes up to 45 mg.
The data show that mature leaves gain weight faster in those first three days than young ones, since the young ones only grew 0.12 mg over the course of three days, while the mature ones grew approximately 5 mg.

(b)Using our interpolator to estimate the largest mean weights for larvae taken from mature leaves and young leaves, this is the output:
Largest mean weight of young leaves: 45.6921 on day: 10.5
Largest mean weight of mature leaves: 19.1740 on day: 9.5

The young leaf peak at approximately 45 mg will be significantly higher than the mature leaf peak approximately 19 mg; however, the spline shows that the larvae weights peak very close to the 10-day mark. For young larvae, the peak is on day 9.5, suggesting that there is some room to grow a bit more before they start to lose weight. The data are consistent with the hypothesis that tannins in mature leaves inhibit growth. The maximum weight for larvae on mature leaves (19.2 mg) is significantly lower than that of young leaves (45.7 mg), therefore the presence of tannins in the mature leaves clearly inhibits their ability to gain mass.

Exercise 7

For question 7, we used data from the Observed Timeseries of Annual Average Mean Surface Air Temperature in Greenland from 1901 to 2024 from here. The data compare the real temperature values over the course of those 123 years to the equivalent normalized (Gaussian) smooth temperatures. Understanding climate change is crucial to understanding the behavior of climate in general, whether short-term or long-term behavior, and most importantly, how reversible or irreversible it is. We downloaded a csv file of the data that had 3 columns of information. We only used the first two, which included the years and the recorded mean temperatures, to apply our interpolation.
Figure 1 is the plot with the estimated values with a line connecting them.

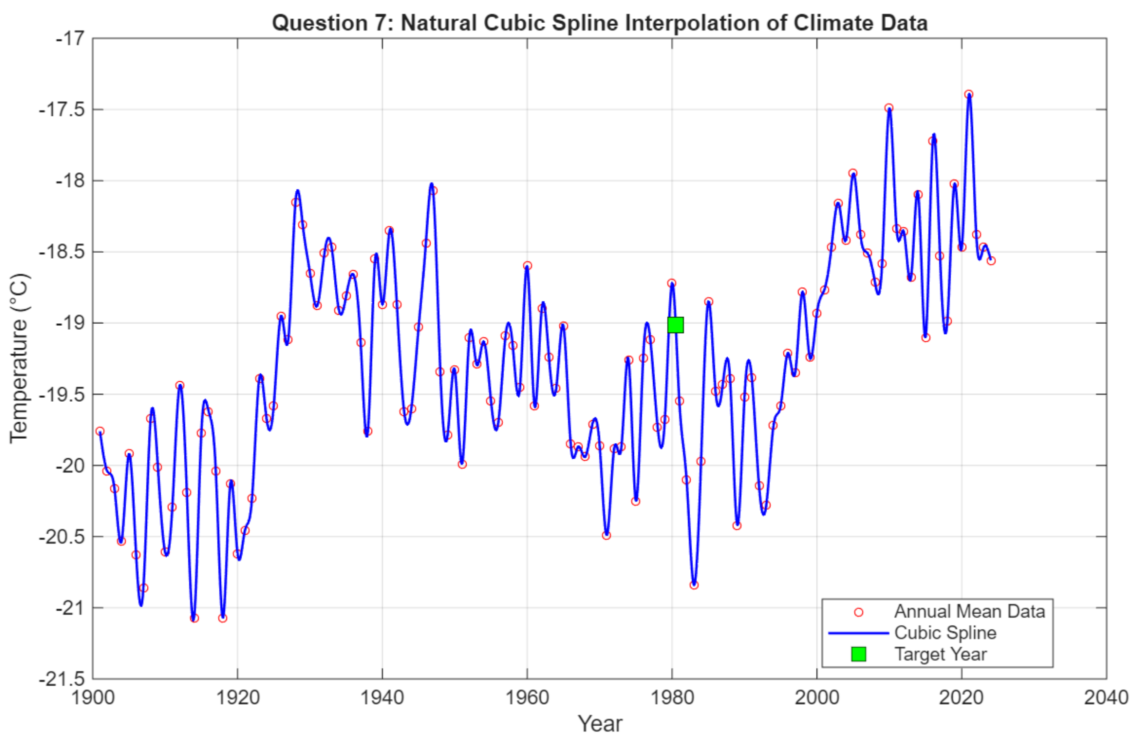

Figure 1: Interpolation of Climate Data

The interpolation fully captures the spikes and differences in the temperatures. This means that the cubic spline with the natural boundaries is very sensitive to capturing extremes. Locally maximum or minimum temperatures are captured by creating a dip to reach them before recovering again to the more frequent/average temperatures. At 1985.5, which is the target year, the interpolated value falls between a significantly colder period.

Exercise 8

For question 8, we used the same data but only every other data point, compared the estimated values to the actual data points we did not use, and lastly, we used every third point. Below in Figure 2 and Figure 3 we see the results of these interpolations.

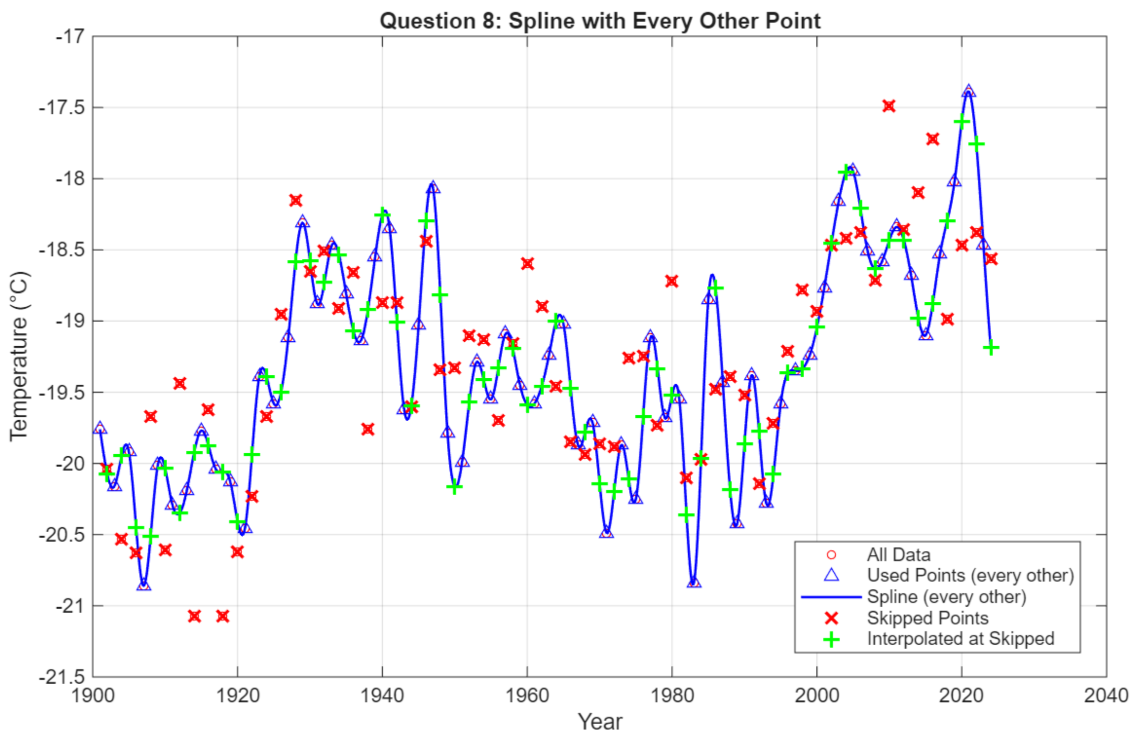

Figure 2: Comparison of interpolation data with every other point

Subsampling the data shows us how sensitive the cubic spline interpolation is to the density of the spacing between intervals. When every other point is used, the spline remains pretty stable and covers most of the points, and the errors at the skipped points tend to be small, as seen in Figure 2. When every third point is used, the spline is not accurate enough to reconstruct the information lost between the points not taken into account, as seen in Figure 3. The errors also increased reflecting the major difference between the spline that passes through all the points and every third one. It mainly misses the local extremes in temperature, while it manages to cover most of the average temperatures, maintaining a smooth line. Therefore, the results suggest that the density of interpolation points matters especially in the regions where the data are most dynamic.

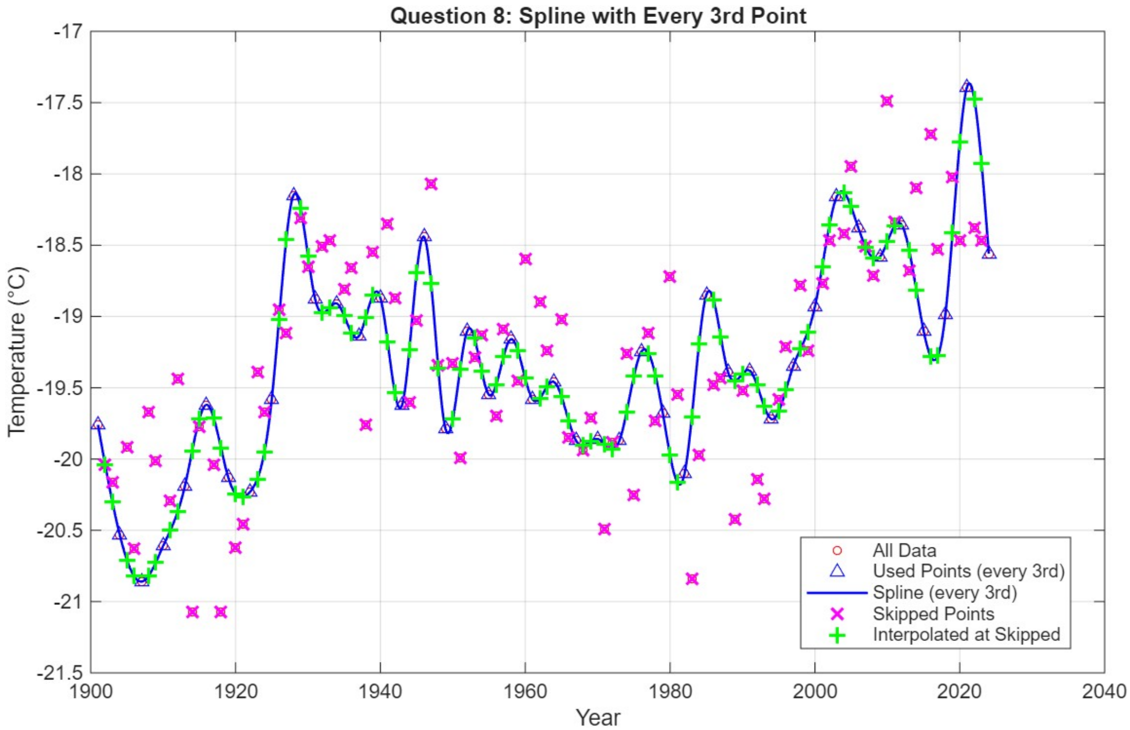

Figure 3: Comparison of interpolation data with every 3rd point

Sources

https://climateknowledgeportal.worldbank.org/country/greenland/climate-data-historical https://www.mathworks.com/matlabcentral/answers/1970399-read-csv-data-with-numbers-characters-and-text